Fiscal Year:	FY 2020	**Task Last Updated:**	FY 11/13/2020
PI Name:	DeChurch, Leslie Ph.D.		
Project Title:	SCALE: Shared Cognitive Architectures for Long-term Exploration		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Behavior and performance		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HFBP**:Human Factors & Behavioral Performance (IRP Rev H)		
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Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:	NOTE: Previously at Georgia Institute of Technology until July 2016.		
Project Type:	Ground	**Solicitation / Funding Source:**	2014-15 HERO NNJ14ZSA001N-Crew Health (FLAGSHIP & NSBRI)
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No. of PhD Candidates:	5	**No. of Master' Degrees:**	0
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No. of Bachelor's Candidates:	4	**Monitoring Center:**	NASA JSC
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 6/30/2021 per NSSC information (Ed., 10/21/20)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Contractor, Noshir Ph.D. (Northwestern University) Johnson, Jeffrey Ph.D. (University of Florida, Gainesville)		
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Task Description:	NOTE: Continuation of "SCALE: Shared Cognitive Architectures for Long-term Exploration," grant NNX15AM26G, due to Principal Investigator move to Northwestern University from Georgia Tech in Fall 2016. Among the remarkable team challenges NASA faces in long distance space exploration (LDSE) missions is the need to maintain team shared mental models (SMMs). Maintaining team SMMs requires the ability to detect shifts in cognition that will likely occur during the mission that could lead to ineffective crew functioning and performance. Maintaining team SMMs also requires validated countermeasures for bringing team members' cognitive understanding back into alignment. Leaving low Earth orbit is extreme teamwork -- team SMMs need to be maintained within teams operating close up (the crew), and between teams operating at an unprecedented distance (i.e., the crew & ground; 33 million miles in the case of a Mars Mission). A multidisciplinary research team will leverage expertise in Psychology, Industrial Engineering, & Anthropology to understand the emergence and outcomes of critical shifts in team cognition over LDSE missions. What are the triggering events of SMM divergence, how can we detect them, and which countermeasures most effectively bring them back into alignment? This project leverages a novel conceptual framework of shared cognitive architecture (SCA) to understand the patterns of SMMs that dynamically link members of teams, and teams to other teams, as they go beyond low Earth orbit. We use semantic analysis to identify cognitive shifts, and relational event network analysis to understand the antecedents and consequences of these shifts. We use these alongside an agent-based model fit on LDSE analogue data, so that we can explore an exhaustive set of potential triggering conditions that must be unpacked to conduct efficient ground analogue research. We then conduct this research in HERA (Human Exploration Research Analog), Moonwalk, and Antarctica. The project culminates in the evaluation of a dashboard fed with the results of computational modeling, human validation, and lexical markers to detect and suggest countermeasures to maintain SMMs through time and space.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Many of the environmental conditions faced by crews during LDSE missions are not unique to space travel. With our research program, we hope to reveal how conditions seen in organizations across the world impact shared cognitive architecture, and what we can do to mitigate these risks. Our meta-analysis has begun to investigate the impact of shared cognitive architecture on team outcomes across a variety of environmental conditions seen in LDSE missions that can also be applied to work in all organizations.
	Over the course of this five-years programmatic research program, we have three major foci of activity: (1) to develop and test ways to measure shared cognition in teams and multiteam systems, (2) to explore the antecedents, consequences, and moderating conditions associated with shared cognition, and (3) to model shared cognition in teams and within multiteam systems so as to inform interventions. In Year 2, we made significant strides in each of these areas with six major projects that speak to one or more of the above foci. In Study 1, we used meta-analysis to explore the conditions under which shared cognition is most important to crew performance. We explored three types of moderating variables, including team type (e.g., team diversity, autonomy, and lifespan), team context (e.g., virtuality, temporal and geographic dispersion, and organizational embeddedness), and team conditions (e.g., physical and psychological stressors). Results helped us determine (1) the extent to which prior teams research is relevant to long distance space exploration, (2) factors which need to be modeled effectively in order to predict, diagnose, and prevent breakdowns in shared cognition, and (3) which antecedent and moderating conditions are most relevant to cognitive sharedness in long distance space exploration. In Study 2, we developed an approach to using text analytics as an unobtrusive means of measuring shared cognition using the archived mission transcripts from the Skylab missions. We identified the text-based signatures of cognitive breakdown that ultimately led to conflict in Skylab 4. We also explored the dynamic conversational patterns in each mission that were indicative of cognitive sharedness or its breakdown. This study informs our broader goal of developing efficient and unobtrusive measures of shared cognition in multiteam systems. In Year 3, we expanded the analysis of Study 2 to examine the reasons why the cognitive breakdown was not averted earlier. We refined our previous analysis to look at the specific subject areas where understandings diverged and the characteristics of these subjects. Our findings suggest that discussion of scheduling, morale, and stressors were highly constrained by the medium and norms of communication and this constraint hindered the development of shared understandings around these issues. In Study 3, we developed an agent based model (ABM) of crew member mental models and estimated parameters using empirical data collected from the HERA analog. We developed the ABM in two phases. First, we canvassed the relevant literature on shared cognition to identify its antecedent conditions. Second, we collected laboratory and analog data to estimate and then test the parameters of the agent based model. In particular, we included personality factors, social relations, situational factors, and task stream characteristics that associate with cognitive sharedness. Then, we implemented the model in the NetLogo ABM platform. The agent-based model estimates the probability that the individual i mental model changes from time t to time t+1. Then, we used BehaviorSearch tool to estimate the parameters that best fit the computational model to the empirical data. In Years 4 and 5 we continued our work on Study 3 (ABM) by conducting virtual experiments to investigate how leadership network structures affect the development of shared mental models. This study speaks directly to our major goal to model shared cognition in teams to inform interventions. We use data from four crews of 4 living in a NASA space analog for 45 days to develop an agent-based model and a follow-on “what-if” computational experiment in which we manipulate leadership structures. We built an agent-based model to explicate how leadership network structures affect mental model development in teams over time. Specifically, we developed a computational model operationalizing shared mental models as network ties among crew members. The strength of ties represents the degree to which each pair of crew members holds a similar understanding of the tasks or the team. The computational model considers aspects of the individuals, their relations with one another, the tasks they complete, and the nature of their experiences and interactions as the mission unfolds to continuously update the degree to which they share mental models. We parameterized the model using data from the four crews. We then used the model to conduct “what if” computational experiments on teams manipulating the leadership network structures and observing their effects on shared mental models. Fitting the empirical data to the computational model shows in addition to leadership, individual cognitive ability, team identity, and negative ties are consistent predictors of shared mental models. Results of computational experiments support the proposition that shared leadership results in the most shared mental models, followed by hierarchical leadership, and then connected leadership; factionalized and disenfranchised leadership structures result in the least shared mental models. The insight that hierarchical leadership results in a more similar shared mental model than connected leadership is especially noteworthy and perhaps counter-intuitive.

Task Progress:	Studies 4-5 speak directly to our major goal of identifying antecedents and consequences conditions of cognitive sharedness. In Study 4, we explored how extended periods of isolation and confinement influence team cognition. We conducted two empirical studies examining team cognition over the course of the space analogue missions. In these studies, we explored the development and impact of three aspects of team cognition in analog space crews in order to understand their natural fluctuations, and how these fluctuations are affected by communication delay, and extended isolation. Finally, we investigated how these different types of team cognition differentially impact various types of team performance. Results show that team cognitive similarity has opposite effects on different dimensions of performance -- similarity benefits execute performance but harms creative thinking. Second, whereas the mean team similarity predicts creative performance, the trend in similarity is more predictive than the mean for problem solving performance. This reveals that the relationship between team cognition and performance is more nuanced than previously conceptualized, and suggests future research is needed to examine team cognition as a dynamic construct. In Study 5, we used Project RED relay data to understand the nature of communication errors in large, distributed, space exploration systems. Using a network routing task, we collected data from NASA's Human Exploration Research Analog (HERA) Campaign 3, 4, and 5 as well as university students. For HERA, we studied 12-person networks consisting of 13, 4-member crews, each interacting with 31 different 8-member mission control centers. In total, we had 299 participants. Participants engaged in a network routing task in which each individual is required to choose 2-3 direct contacts in order to relay messages that must reach specific others in the network in the fewest number of steps. From this network routing task, we measured network acuity based on the extent to which each individual routed messages through their contact who was in fact on the shortest path to the final target. Results show that network acuity among HERA crews was significantly higher than MS members in Campaign 4 and 5, but not Campaign 3. Additionally, we found communication delays did not impact network acuity among HERA crews, but reduced acuity among MS members. Further, crew members who scored lower on the personality characteristic of conscientiousness had higher network acuity in both Campaigns 3 and 4. Finally, crews' network acuity was associated with personality characteristics of openness to experience, agreeableness, and neuroticism in Campaign 4, but not Campaign 3. Overall, our findings suggest that selecting crews with high network acuity will play a key role in alleviating the risk of information-sharing failures within a multiteam system under conditions of communication delays. This research will be useful for investigating shared cognition in any modern-day organization facing complex collaborative challenges, such as NASA space exploration, large scientific consortia (e.g., CERN--European Organization for Nuclear Research), cybersecurity teams, healthcare systems, and the military. Furthermore, findings could then be leveraged to develop system-wide interventions that increase overall work efficiency and resilience in safety-critical systems. In this review period, we have collected data across multiple platforms to begin to understand the antecedent, consequent, and moderating conditions associated with shared cognition. In addition to providing Earth Benefits in the area of shared cognition in teams and multiteam systems, we are also testing our hypotheses in analogs that recreate the unique contexts in which astronauts operate (i.e., extreme, isolated/confined environments). We will compare results across research paradigms to see how effects of isolation/confinement may moderate shared cognition's role in team functioning, thus providing unique findings that will inform NASA's operations aboard the International Space Station (ISS) and on Long-Duration Space Exploration (LDSE) missions. In this review period, we have collected data across multiple platforms: HERA analog, Russia's IBMP Ground-based Experimental Complex (NEK (Nezemnyy Eksperimental'nyy Kompleks)), and Antarctica. (1) This last year we tracked shared cognition and team performance over time in Campaign 5 of NASA's Human Exploration Research Analog (HERA) Campaign. Each of the HERA campaigns has a unique research emphasis and the emphasis of Campaign 5 is to better mimic living quarters of future vehicles traveling beyond low Earth orbit. (2) In addition to the research being conducted out of HERA in the past year, we designed and launched a sister study located in Antarctica. Participants for this study were required to be astronaut-like and required to be in some form of a work-team or work-group. The combination of the isolating, confined, and extreme environment of the Antarctic winter created an ideal place for a space exploration analog study. This study is designed to investigate team dynamics, specifically shared cognition, in isolated, confined, and extreme environments and how these dynamics occur and change over long periods of time. The study surveyed the experience of participants as they worked in their teams over the course of six months during their stay in Antarctica throughout the austral winter and will be drawing to a close within the next few months. (3) In addition to the research being conducted out of HERA in the past year, we also collected team cognition data in NEK / SIRIUS (Scientific International Research In a Unique terrestrial Station) 2019. We conducted a thematic analysis of space crew exit interviews from 11 individuals recounting their experiences (5 ISS astronauts and the 6-member SIRIUS '19 crew).
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